NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment : 01

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Final Year Project Progress Report

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1. Introduction

Respiratory diseases such as asthma, COPD, bronchitis, and pneumonia often remain undiagnosed in early stages due to mild symptoms, limited access to specialists, and expensive clinical tests. BreatheWell is an AI-powered web application designed to assist in early detection and continuous monitoring of lung diseases by analyzing cough and breathing sounds along with user-reported symptoms. The system also provides personalized care plans, environmental alerts, and health tracking features.

2. Project Objectives

The main objectives of the BreatheWell project are:

- To detect lung diseases using AI-based cough and breathing sound analysis
- To provide a non-invasive and accessible respiratory health monitoring solution
- To enable symptom tracking and early warnings for disease progression
- To offer personalized care plans and exercise recommendations
- To integrate environmental air quality alerts for preventive care

3. Work Completed (40% Progress)

3.1 Requirement Analysis

- Functional and non-functional requirements were identified and documented
- User roles (patient and doctor) were defined
- Use case diagrams and detailed use case descriptions were prepared

3.2 System Design & Architecture

- A Three-Tier Architecture was designed (Frontend, Backend, Database)
- UML diagrams including Class, ER, Sequence, and Activity diagrams were created
- 4+1 architectural view model was documented

3.3 UI/UX Design

- Application screens were designed using Figma
- Interfaces designed include Login, Registration, Dashboard, Symptom Submission, Cough Recording, Reports, Care Plan, and Environmental Alerts

3.4 Frontend Development (Partial)

- React.js project setup completed
- Login, Registration, Dashboard layout, and Symptom Submission screens implemented
- Basic navigation and component structure developed

3.5 Backend Development (Initial Phase)

- Flask backend structure created
- User authentication (Signup/Login) implemented

- MongoDB database connection established
- Initial API endpoints developed

3.6 AI & Machine Learning Research

- Research conducted on respiratory audio datasets
- Studied feature extraction techniques (MFCC, spectrograms)
- Defined AI workflow for cough and breathing analysis

4. Tools & Technologies Used

- **Frontend:** React.js
- **Backend:** Flask / Node.js
- **Database:** MongoDB
- **Authentication:** Firebase
- **AI/ML:** Python, Deep Learning (planned)
- **Design:** Figma

5. Methodology

The **Waterfall Model** is adopted for this project due to clear requirements and academic project constraints. Completed phases include Requirement Analysis and System Design, while implementation is currently in progress.

6. Challenges Faced

- Availability of reliable cough and breathing sound datasets
- Ensuring clean and noise-free audio input
- Integrating AI models with the web backend

7. Future Work

- Development and training of AI models for audio classification
- Completion of backend logic for smart diagnosis and care plans
- Integration of environmental alert APIs and report generation
- Frontend enhancements and system testing

8. Conclusion

At the 40% completion stage, BreatheWell has successfully completed requirement analysis, system design, UI/UX design, and partial frontend and backend implementation. The foundation for AI-based respiratory disease detection has been established, and the remaining work will focus on AI integration, full system implementation, and testing.